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Native GeoArrow Computation Engine in R

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Executive Summary

This proposal seeks to ameliorate three challenges facing the R-spatial ecosystem.

First, spatial analysis in R “is slow” and can be dramatically improved. The de facto package for vector-based spatial analysis in R is the `{sf}` package. Due to its ergonomic API and its integration with the tidyverse, it is the common choice for spatial data analysis. However, `{sf}` can struggle to scale effectively and has caused developers to turn to other tools and other languages.

Secondly, the past few years have seen the rapid adoption of Apache Arrow—a standard for in-memory representation of arrays and tables. The spatial ecosystem, in particular, has been quick to adopt GeoArrow, an extension to the Arrow standard that defines how spatial data can be represented with Apache Arrow.

The standardization on (Geo)Arrow enables the development of packages that can work on in-memory data without incurring any additional overhead costs of deserializing and serializing data into intermediate objects.

Lastly, concurrent to the development of GeoArrow has been the development of GeoParquet. At present, GeoParquet support in R has focused on consuming the format. To my knowledge, there is no R package that provides tailored support for writing GeoParquet as well as reading from remote storage.

I seek \$5,000 USD to fund the completion of the development of a new Rust-based, GeoArrow native, and highly performant spatial computation engine for R called `{geoarrows}`. The R package will provide:

- GeoArrow-native and parallelized computation engine
- Persistent spatial RTree and KDTree spatial indexes for efficient spatial querying and joining
- Drop in compatibility with `{sf}`
- Enhanced GeoParquet support such as spatial sorting, bounding boxes on row-chunks for predicate pushdown, and persisting coordinate reference systems.

Signatories

Project team

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Contributors

Consulted

The Problem

The (de)serialization problem

While `{sf}` is a foundational package for the R ecosystem, its performance has failed to scale effectively as the size of data that people commonly work with has grown. This is a common experience. Users [report slow distance calculations on modest data](#), [write their own batching workarounds to finish jobs at all](#), and [performance](#) remains an open topic in `{sf}`'s own issue tracker.

The biggest challenge with `{sf}` is primarily the fact that geometries are stored in R as a list. Each operation that `{sf}` performs has to be first converted to a GEOS or S2 geometry. The S2 or GEOS libraries perform operations on their native geometries and then `{sf}` captures their results and again converts them back to their R representation. This limitation has led GDAL to create a GeoArrow native “column-oriented” reader which uses Apache Arrow natively to improve performance. This process of converting to and from is the expensive (de)serialization that the GeoArrow standard exists to partially help address.

GeoArrow as a standard exists to address exactly this problem. It has been a focus of [Spatial Data Science Across Languages](#) since its inception in 2024 and [again in 2025](#), and as a result we are seeing coordinated adoption across all languages. As an indicator of its importance, GDAL added a [column-oriented Arrow API](#). GeoPandas, too, added [native GeoArrow support](#).

At present, R support is limited. The R package `{geoarrow}` provides a `{nanoarrow}` and `{arrow}` compatible vector representation. The ISC-funded `{geoarrowWidget}` package assists in transporting `{geoarrow}` vectors to the browser. However, the R community lacks a GeoArrow-native compute engine that can scale effectively.

Alternative Computation Engines

`{geos}` has historically been the fastest R package in the ecosystem but is very low level and thus not very widely used.

Currently, there are two primary options for scalable geospatial computation engines in R: 1) `{duckspatial}` and 2) `{sedonadb}`. Both of these face similar issues: neither work with native `data.frames` and both require external downloads. `sedonadb` is especially challenged due to not being on CRAN.

The `duckspatial` R package uses the [spatial extension](#) to DuckDB. DuckDB relies on downloading and installing an extension. The size alone is reason for concern. Secondly, it requires [static linking](#). `{sedonadb}` provides bindings to [Apache Sedona](#). The most fit for large scale analysis but is not available on CRAN and lacks sufficient documentation as well as has a bespoke API. The latter, while arguably the most scalable package, is not available on CRAN and lacks sufficient documentation leading to limited adoption.

GeoParquet

- GeoParquet is becoming the standard for storing and sharing spatial data, particularly on cloud object storage
- The {geoarrow} R package provides GeoParquet metadata to the {arrow} R package so that {arrow} can read GeoParquet natively.
- Writing GeoParquet is can be done via `arrow::write_parquet()` (note that sfarrow delegates to arrow directly)
- This is minimally viable GeoParquet.
- Parquet's benefit is that row-group metadata can be used to efficiently filter data.
- At present there is no way easy way to record spatial metadata in a GeoParquet file
- Geometries are written in the order that they are in the data.frames
- Spatial proximity is entirely ignored and thus row-groups are not are not spatially organized meaning libraries cannot spatially filter data without ingesting it.
- lastly, there is no projjson support in any existing library which means that the CRS of our data can be dropped and downstream tools will not know its CRS
- The result is a file that other tools can open but that has lost its CRS, and that forces a full scan to answer a spatial query — which on remote storage means transferring the entire file

we need an R package with a modest footprint, no system dependencies, no external downloads, that works on GeoArrow natively, provides fast, scalable, and efficient spaital operations, as well as true GeoParquet support.

The proposal

This proposal is to fund the completion of the R package, {geoarrows}, which is a geoarrow-native compute engine for the R ecosystem.

The R package will provide memory efficient and parallelized implementation for geometric operations out of the box. In addition, it will provide advanced spatial join and query operations. The R package will integrate into the existing ecosystem with a focus on integration with {arrow}, {dplyr}, and {sf} to minimize friction in adoption.

This proposal is for work in the following

1. Core geometry algorithm implementation
2. Parallelization and
3. {sf} integration

Overview

The {geoarrows} package is a GeoArrow-native spatial computation engine designed for ergonomic use within the R ecosystem. The package is in the middle of its development life cycle and this grant would fund its completion.

This package aims to be faster than geos, competitive with duckspatial and sedonadb, and as ergonomic as {sf} (with a drop in replacement via `mask_sf()`) for **in memory** computation.

Today, the package provides:

- performant readers for shapfile, GeoJson, and FlatGeobuf
- R-tree and KD-tree spatial indexes

- Spatial, nearest neighbor, and distance band joins
- User defined functions registered with `{arrow}`

Completion of the package is split into four milestones:

1. Complete core geometry engine so that every algorithm provided by the Rust crate `geo` as of this writing is available from R. This would achieve near parity with `{sf}` on most operations.
2. Create spatial indexes that can be written to disk which can be used to spatially sort geometries for more efficient GeoParquet storage.
3. Provide drop in `{sf}` compatibility via `mask_sf()` so that `{georowr}` can be used as a backend for `{sf}` and release the package on CRAN.
4. Persist coordinate reference systems, so that GeoArrow transfer between R, Python, DuckDB, and GDAL can be lossless. This milestone also targets GeoParquet column meta persistence enabled by milestone 2.

Prior art

The `{georowr}` package is the culmination of years of exploring the intersection of Rust, geospatial, and GeoArrow. Initially, the `{rsgeo}` package was created to be a zero-dependency and high-performance spatial computation engine with Rust based conversion to and from `{sf}`.

During this time GeoArrow began to pick up. With the `{anime}` line matching algorithm being a proof of concept for use of GeoArrow in R built on my `arrow-extendr` Rust library which provides FFI between R and Rust for `{arrow}`, `{nanoarrow}`, `{polars}`, and `{geoarrow}` packages.

In 2024, the initial version of this package was started. But development was paused due to an unstable `georowr-rust` API, and challenges with Arrow FFI that are now resolved.

Detail

- Core R package:
 - File readers:
 - * `flatgeobuf`
 - * `shapefile`
 - * `geojson`
 - Spatial Indexes:
 - * KD-tree
 - * RTree
 - Spatial Sorting:
 - * Hilbert
 - * Sort-tile-recursive
 - Spatial Joins
 - * All basic spatial predicates
 - * KNN join
 - * Distance Joins
 - Full coverage of all geometric operations in `georust/geo`
- `{arrow}` R package registered user defined functions
- `sf` compatibility S3 method registration
- CRS propagation?

Readers

- provided by geoarrow for geosjons and flatgeobuf
- shapefile needs manual im

Algorithm Implementation

- Relying on geo for core geometric operation algorithms
- Joins are hand crafted used both geo and geo-index for creating query trees

Minimum Viable Product

Architecture

- Rust based R package via [extendr](#)
- Arrow FFI between R and Rust via [arrow-extendr](#)
- GeoArrow array representation via [geoarrow-rs](#)
- Native algorithms using [geo](#)
- Coordinate reference systems via [projjson](#)
- Spatial Indexing using geo-index
- Readers and writer using:
 - geozero <https://docs.rs/crate/geozero/0.14.0>
 - geojson
 - flatgeobuf
- Parallelism via <https://docs.rs/rayon>

Assumptions

External dependencies

Limitations

- No CRS support at present—addressed by Milestone 4 via the projjson crate
- Limited spherical / geodesic support—true in duckspatial, too

Project plan

Start-up phase

Technical delivery

- Core Algorithm Implementation
- Spatial Joins
- Parallelization and optimization
- {arrow} registered
- pkgdown site
- spatialbench benchmark
- duckspatial benchmark
- CRAN release

Other aspects

- i'd like to submit the package to rOpenSci or R-Spatial if they'd take it
- Social media posts
- Joint Blog post with R Consortium

Budget & funding plan

I request \$5,000 USD to fund the development of {geoarrowr}.

Milestone 1 - Complete Core Geometry Engine Implementation (\$1,500)

This milestone funds the development effort to complete the core computation engine in {geoarrowr} such that there is near-complete parity with {sf}. To accomplish the milestone, the following will be completed:

- Geometry validation: `st_is_valid()`
- Geometry reparation: `st_make_valid()`
- Geometry type checking: `st_geometry_type()` and `st_is()`
- Collection extraction: `st_collection_extract()`
- Perimeter calculation: `st_perimeter()`
- Coordinate comparison: `st_equals_exact()`
- Jittering: `st_jitter()`
- Node: `st_reverse()`, `st_node()`
- Areal interpolation: `st_interpolate_area()`
- Bounding box: `st_bbox()`
- Polygon constructors:

The following features are out of scope of this proposal as there are no pure rust alternatives to rely on at present.

- 3D geometries
- Snapping
- Polygonization
- Geometry Projection

Milestone 2 - Persistent Spatial Indexes and Spatial Sorting (\$1,000)

This milestone funds the development of creating and persisting spatial RTree and KDTree spatial indexes. These spatial indexes are created on the fly for each operation performed by {sf} as well as other libraries. For operations that require repeated look ups, persisting the index can be highly efficient.

Further, these indexes can be used to perform sorting of geometries to ensure that nearby geometries are ordered together. Spatial sorting can greatly improve the effectiveness of row group statistics in GeoParquet files. Partial reads of a GeoParquet file can leverage the sorting more effectively.

For spatial sorting, Hilbert and sort-tile-recursive (STR) will both be supported. {geos} only provides STR sorting and spatial RTrees, to my knowledge, do not have support in the R ecosystem.

Milestone 3 - {sf} Compatibility and CRAN release (\$1,500)

In order for the package to be useful to the broadest community, this milestone will create drop in support for {sf} via S3 namespace masking. Users can call `mask_sf()` to overload the {sf} namespace with generics that enable {sf} to work with {geoarrowr} geometries.

Following, documentation, examples, and benchmarks will be tidied up, and rewritten for ease of adoption.

One **risk** is that the source tarball is 14mb which is 9mb larger than the requested CRAN limit. However, exceptions can be made. For example, the prior art `{rsgeo}` was a [9mb tarball](#). And other Rust based R packages such as `{fcl}`, are [15mb](#) and supported on CRAN.

One **limitation** is that the `geo` crate that we rely on geometry operations does not have full spherical support whereas `{sf}` does via the `{s2}` package. In the event that a spherical CRS is used

Milestone 4 - GeoArrow and GeoParquet metadata

At present, `geoarrowrs` cannot propagate a coordinate reference system (CRS) across the R to Rust FFI boundary. Milestone 3 is possible by leaning on `{sf}`'s CRS object and adding it as an attribute on the `geoarrow` vectors which requires PROJ to be installed. This milestone would ensure that no external system dependencies would be needed.

Further, the GeoArrow specification records CRS as column level metadata in the `projjson` format. To support this, the `projjson-rs` crate was created which provides strongly typed reading and writing of `projjson` with no external system dependencies.

As a result, GeoArrow transferred from DuckDB, R, Python, and GDAL would propagate the CRS correctly.

The column level metadata is also utilized by GeoParquet. As such, this milestone completely implements the GeoParquet column metadata including:

- encoding
- geometry_types
- crs
- bbox
- orientation
- edges
- epoch

Notably, the `edges` indicates wh

Success

Definition of done

Measuring success

Correctness tests with `sf`

Future work